

# THE TURNING NUMBER OF A PLANAR CLOSED CURVE AND HEIGHT FUNCTIONS

ABSTRACT. For a closed planar curve, we show how to compute its turning number (i.e. the winding number of its tangent field) in terms of the sings of the curve's curvature at the critical heights of the curve to a given line (see Figure 1)

## 1. CURVATURE AND THE TURNING NUMBER

1.1. **Curvature.** We will describe a closed curve in the plane as a regular periodic map  $p : \mathbb{R} \rightarrow \mathbb{C}$  of class  $C^2$ . We will assume that  $p$  is parametrized by arc-length, so that  $|\dot{p}(t)| = 1$  for all  $t \in \mathbb{R}$ . The periodicity assumption implies the existence of a number  $\tau \in \mathbb{R}$  such that  $p(s + \tau) = p(s)$  for all  $s \in \mathbb{R}$ . In this case the curve is

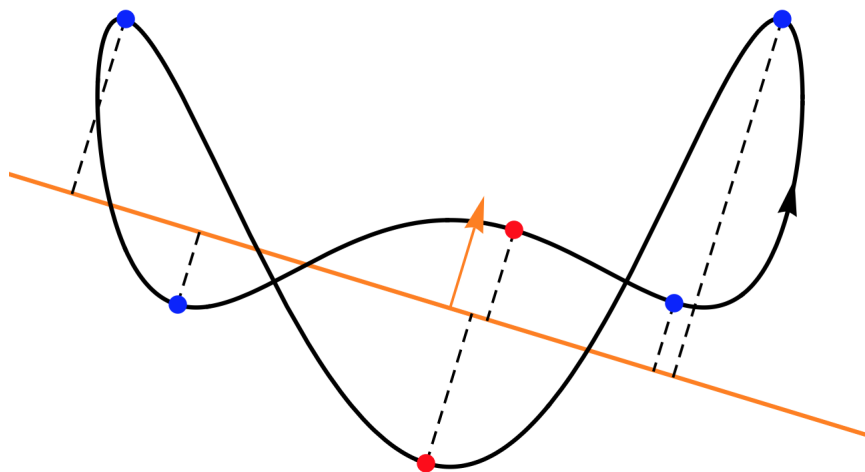


FIGURE 1. Critical heights of a curve (in black) to a line (in orange). At the red critical points the curve has negative curvature and at the blue points it has positive curvature. The turning number of the curve is obtained by adding the total number of blue dots, subtracting the total number of red dots, and dividing the result by two. Observe that at the curve sections joining points of different color the tangent field has total rotation equal to zero, and at the sections joining points of the same color the total rotations is  $\pm\pi$  (the plus and minus sign depending on whether the points are blue or red, respectively).

completely determined by its values on the interval  $[0, \tau)$ . The tangent and normal fields to the curve are

$$T = \dot{p} \text{ and } N = iT$$

Differentiating the formula

$$\langle \dot{p}, \dot{p} \rangle = 1$$

we obtain

$$\langle \ddot{p}, T \rangle = 0,$$

so that

$$\ddot{p} = \kappa N,$$

for a periodic function  $\kappa : \mathbb{R} \rightarrow \mathbb{R}$ . This function can be computed explicitly as

$$\kappa = \langle \ddot{p}, N \rangle = \langle \dot{T}, N \rangle.$$

**Definition 1.** The function  $\kappa$  above is known as the *curvature function* of the curve  $p = p(s)$ .

For a more detailed description of the meaning of curvature the reader is referred to standard books on Differential Geometry like [1, 3].

**1.2. The winding number.** Since the tangent field is unitary we can always write

$$(1.1) \quad T(s) = \cos \theta(t) + i \sin \theta(t)$$

It can be shown that if  $p$  is of class  $C^2$  (so that  $T$  is of class  $C^1$ ) then  $\theta : \mathbb{R} \rightarrow \mathbb{R}$  can be chosen to be of class  $C^1$  (see [3, Proposition 5, pg. 22]). The condition  $T(0) = T(\tau)$  implies that

$$\frac{\theta(\tau) - \theta(0)}{2\pi} \in \mathbb{Z}.$$

The above number is known as the winding number of  $T$ , and we will denote it by  $n(T)$ . From formula 1.1 we get

$$\kappa = \langle \dot{T}, N \rangle = \langle \dot{\theta} N, N \rangle = \dot{\theta},$$

and hence

$$n(T) = \frac{1}{2\pi} \int_0^\tau \kappa(s) ds.$$

## 2. STATEMENT AND PROOF OF THE THEOREM

**Theorem 2.** Consider a closed curve in the plane parametrized by arc-length by a regular periodic map  $p : \mathbb{R} \rightarrow \mathbb{R}^2$  of class  $C^2$ . For a given unit vector  $N_0 \in \mathbb{R}^2$  let  $h_0 : \mathbb{R} \rightarrow \mathbb{R}$  be the height function

$$h_0(s) = \langle p(s), N_0 \rangle.$$

If the set of critical points  $C_0$  of  $h_0$  is such that  $\kappa(s) \neq 0$  for all  $s \in C_0$ , then

$$n(T, \vec{0}) = \frac{1}{2} \sum_{s \in C_0} \text{sgn}(\kappa(s)).$$

We prove the theorem as follows. Since

$$\begin{aligned}\frac{dh_0}{ds} &= \langle N_0, \dot{p} \rangle = \langle N_0, T(s) \rangle, \\ \frac{d^2h_0}{ds^2} &= \langle N_0, \ddot{p} \rangle = \kappa \langle N_0, N \rangle.\end{aligned}$$

the set of critical points of  $h_0$  is

$$C_0 = N^{-1}(\{N_0, -N_0\}).$$

and for  $s \in C_0$  we have that

$$\frac{d^2h_0}{dt^2}(s) = \pm \kappa(s).$$

The selection of plus or minus sign in the above formula depends on whether  $N(s)$  points in the same direction as  $N_0$  (plus sign) or in the opposite directions (minus sign). We conclude the following:

- (1) The critical points of  $h_0$  are either strict local maximums or strict local minimums.
- (2) The set  $C_0$  of critical points of  $h_0$  is finite, so that we can write  $C_0 = \{s_1, \dots, s_n\}$  with  $s_i < s_{i+1}$ . Due to the periodicity of  $h_0$  there is no loss of generality in assuming that  $C_0 \subset (a, b)$

To simplify notation, for  $1 \leq k \leq n$  we let

$$T_k = T(s_k), N_k = N(s_k), \kappa_k = \kappa(s_k) \text{ and } \sigma_k = \text{sgn}(\kappa(s_k)),$$

and

$$T_{n+1} = T_1, N_{n+1} = N_1, \kappa_{n+1} = \kappa_1 \text{ and } \sigma_{n+1} = \sigma_1.$$

The above assumptions imply that if  $s_k$  is a minimum then  $s_{k+1}$  is a maximum. Similarly, if  $s_k$  is a maximum the  $s_{k+1}$  is a minimum. Hence the cardinality of  $C_0$  is an even positive integer.

We know that

$$n(T, \vec{0}) = \frac{1}{2\pi} \int_a^b \kappa(s) ds = \frac{1}{2\pi} \sum_{k=1}^n \int_{s_k}^{s_{k+1}} \kappa(s) ds,$$

where for  $i = n$  we have defined

$$\int_{s_n}^{s_{n+1}} \kappa(s) ds = \int_{s_n}^b \kappa(s) ds + \int_a^{s_1} \kappa(s) ds.$$

**Lemma 3.** *Let  $S^1$  be the unit circle in  $\mathbb{C}$ ,  $N \in S^1$  and  $\gamma : [a, b] \rightarrow S^1$  a continuous curve. If  $\langle \gamma(t), N \rangle \in \mathbb{R}^+$  for all  $t \in (a, b)$  and  $\langle \gamma(a), N \rangle = \langle \gamma(b), N \rangle = 0$  then*

$$\frac{\theta(b) - \theta(a)}{2\pi} = \frac{\gamma(b) - \gamma(a)}{4iN}.$$

*Proof.* The conditions imply that

$$\frac{\theta(b) - \theta(a)}{2\pi} \in \left\{0, \frac{\pi}{2}, -\frac{\pi}{2}\right\}.$$

A simple case by case analysis shows that the different possibilities can also be computed by the right hand side of the formula proposed in the lemma.  $\square$

**Corollary 4.** *Under the assumptions of the theorem we have that*

$$\frac{\theta(s_{k+1}) - \theta(s_k)}{2\pi} = \sigma_k \left( \frac{T_{k+1} - T_k}{4iN_k} \right).$$

*Proof.* This follows directly from the lemma, observing that the value of  $\sigma_k$  accounts for the fact that the condition  $\langle \gamma(t), N \rangle \in \mathbb{R}^+$  in some cases becomes  $\langle \gamma(t), N \rangle \in \mathbb{R}^-$ .  $\square$

**Lemma 5.** *Under the assumptions of the theorem we have that*

$$\frac{1}{2\pi} \int_{s_k}^{s_{k+1}} \kappa(s) ds = \frac{1}{4} (\sigma_k + \sigma_{k+1}).$$

*Proof.* Using lemma 4 and the identity  $N_i = -iT_i$  we obtain

$$\begin{aligned} \frac{1}{2\pi} \int_{s_k}^{s_{k+1}} \kappa(s) ds &= \frac{\theta(s_{k+1}) - \theta(s_k)}{2\pi} \\ &= \sigma_k \left( \frac{N_k - N_{k+1}}{4N_k} \right) \\ &= \frac{1}{4} (\sigma_k - \sigma_k(N_{k+1}/N_k)). \end{aligned}$$

The proof will be completed if we show that

$$\sigma_{k+1} = -\sigma_k(N_{k+1}/N_k),$$

or equivalently

$$\frac{\sigma_{k+1}}{\sigma_k} = -\frac{N_{k+1}}{N_k}.$$

We prove this last formula as follows. For  $1 \leq k \leq n$  we define

$$\beta_k = \operatorname{sgn} \left( \frac{d^2 h_0}{ds^2}(s_k) \right)$$

and  $\beta_{n+1} = \beta_1$ . Since the minimums and maximums of  $h_0$  appear in alternate form, we have that  $\beta_{k+1} = -\beta_k$ . Finally, since

$$\sigma_k = \beta_k(N_k/N_0),$$

we obtain

$$\frac{\sigma_{k+1}}{\sigma_k} = \frac{\beta_{k+1}(N_{k+1}/N_0)}{\beta_k(N_k/N_0)} = -\frac{N_{k+1}}{N_k}.$$

$\square$

The proof of the theorem is completed as follows. By using Lemma 5 we have that

$$\begin{aligned} n(T, \vec{0}) &= \sum_{k=1}^n \left( \frac{1}{2\pi} \int_{s_k}^{s_{k+1}} \kappa(s) ds \right) \\ &= \frac{1}{4} \sum_{k=1}^n (\sigma_k + \sigma_{k+1}) \\ &= \frac{1}{2} \sum_{k=1}^n \sigma_k \end{aligned}$$

Finally, recall that

$$\sigma_k = \operatorname{sgn}(\kappa(s_k))$$

where  $C_0 = \{s_1, \dots, s_n\}$  is the set of critical points of  $h_0$ .

*Remark 6.* The result of the Theorem holds when that the critical points of the function  $h_0$  are non-degenerate. This holds for almost all unitary vectors  $N_0 \in \mathbb{R}^2$  (see [2]).

#### REFERENCES

- [1] M.P do Carmo. *Differential Geometry of Curves and Surfaces*. Prentice-Hall, 1976.
- [2] L. Nicolaescu. *An invitation to Morse Theory*. Springer Verlag, 2011.
- [3] M. Spivak. *A Comprehensive Introduction to Differential Geometry, Vol II*, volume 2. Publish or Perish, 1979.